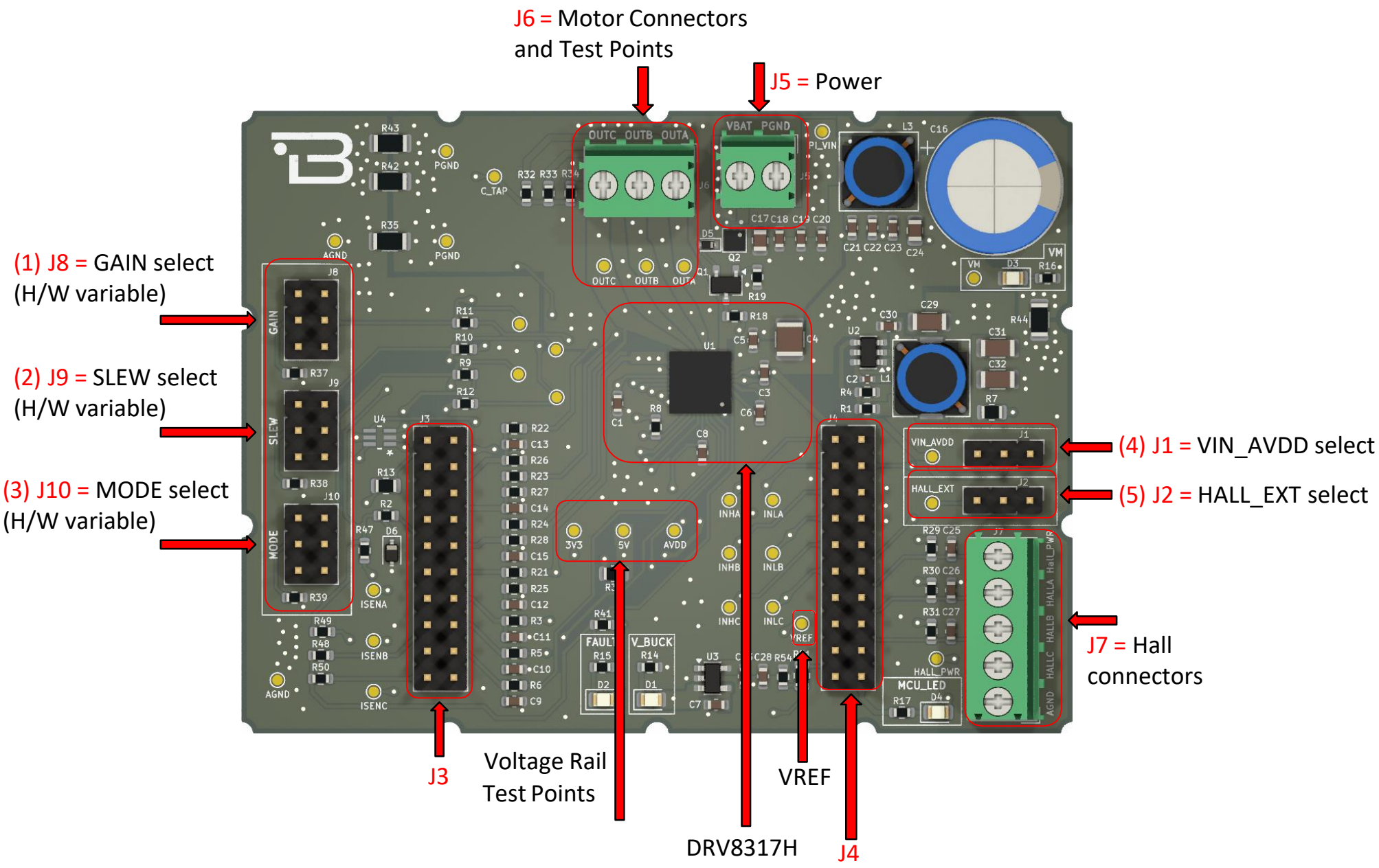
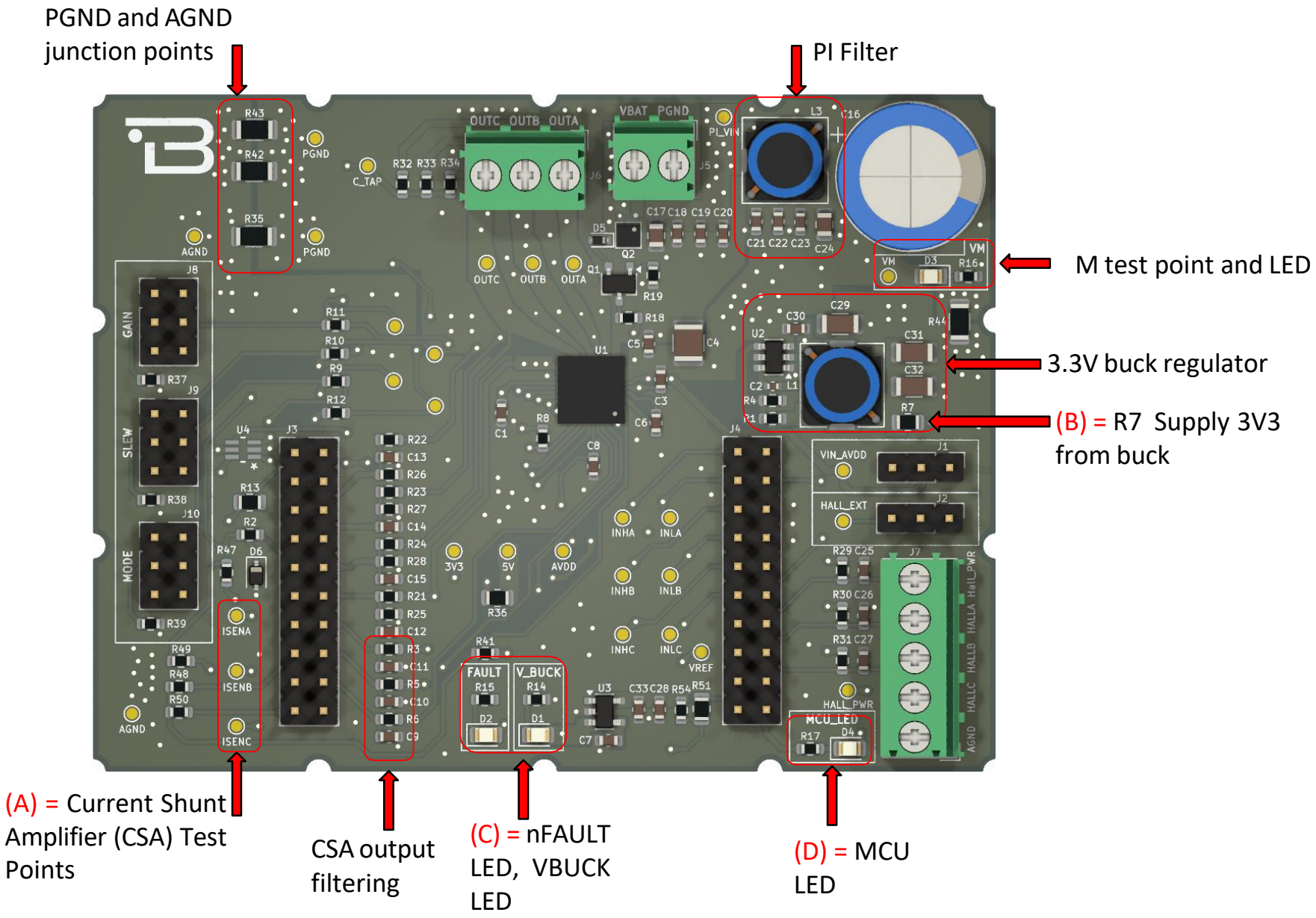
- **Quick Start Guide**

The DRV8317H requires a power supply with a recommended operating range from 4.5V to 20V. To setup and power, follow the sequence below:

- 1. Connect the power supply ground to the PGND of the 2-pin power connector J5 and the power supply positive terminal to the VBAT pin of J5.
- 2. Connect the motor phases to OUTA, OUTB, and OUTC in the correct order. For sensored applications, Make, connect the Hall sensors in the appropriate locations on the 5-pin connector J7. Select 3V3 or Ext on jumper J2 to choose the Hall power source voltage to be either from the 3V3 on the board or from an external supply.
- 3. Apply power by connecting the motor and power connectors to your evaluation board.

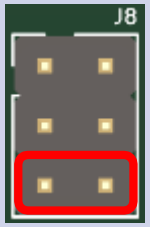
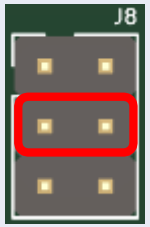
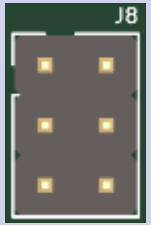
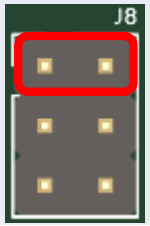

J6 = Motor Connectors and Test Points
J5 = Power
(1) J8 = GAIN select (H/W variable)
(2) J9 = SLEW select (H/W variable)
(3) J10 = MODE select (H/W variable)
(4) J1 = VIN_AVDD select
(5) J2 = HALL_EXT select
J7 = Hall connectors
J3
Voltage Rail Test Points
DRV8317H
VREF
J4


PGND and AGND junction points
PI Filter
M test point and LED
3.3V buck regulator
(B) = R7 Supply 3V3 from buck
(A) = Current Shunt Amplifier (CSA) Test Points
CSA output filtering
(C) = nFAULT LED, VBUCK LED
(D) = MCU LED

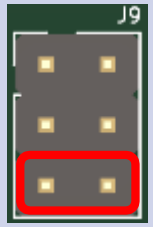
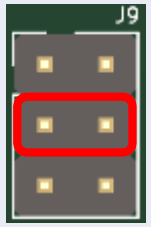
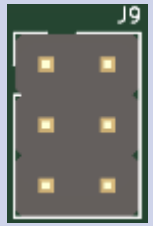
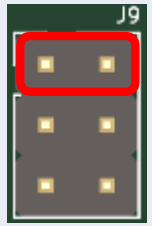
Description of User Selectable Settings on DRV8317H

Id.	Name	Description
A	Current Shunt Amplifier Test Points	Test points to measure the currents of motor phases
B	R7 is populated	Connects buck output to supply 3V3 rail
	R7 is DNP	Supply 3V3 externally from the 3V3 test point
C	V_BUCK	on-board 3.3 V Buck is on
	nFAULT	Lights up when fault condition has occurred on the DRV8317
D	MCU_LED	MCU debugging

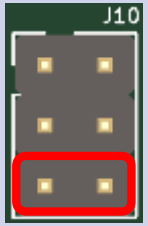
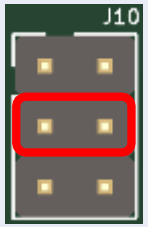
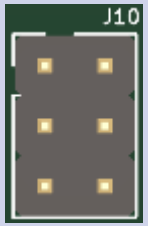
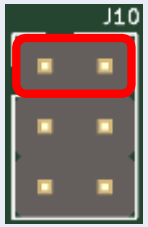
Connections for Header J8 on DRV8317H

Id.	Setting Name	Description	Position	Function
1	Gain Select	Use J8 to select desired gain settings. DNP jumper if DRV8317H is not used.		0.25 V/A
				0.5 V/A
				1 V/A (DEFAULT)
				2 V/A

Connections for Header J9 on DRV8317H

Id.	Setting Name	Description	Position	Function
2	SLEW Select	Use J9 jumper to select desired slew rate settings. DNP jumper if DRV8317H is not used.		25 V/us
				50 V/us
				125 V/us (DEFAULT)
				200 V/us

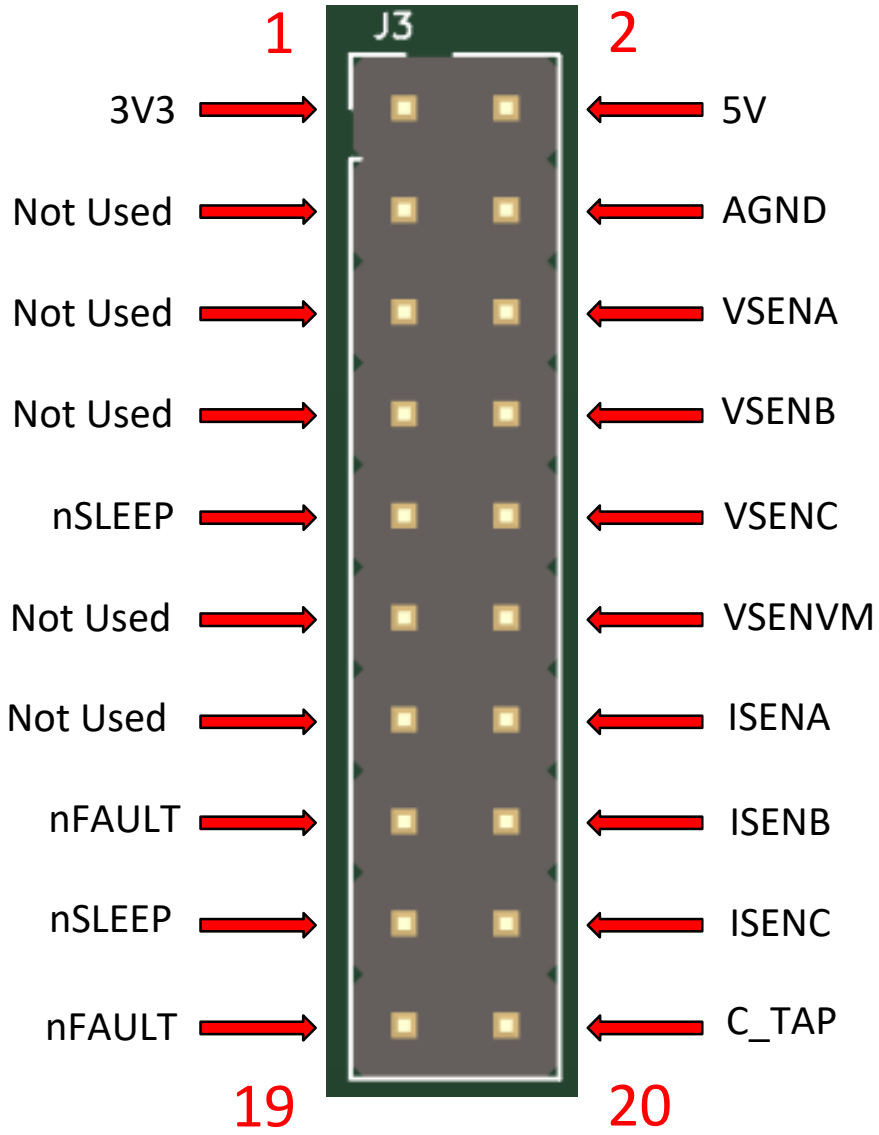
Connections for Header J10 on DRV8317H

Id.	Setting Name	Description	Position	Function
3	MODE Select	Use J10 jumper to select desired PWM mode. DNP jumper if DRV8317H is not used.		6x PWM mode
				6x direct PWM mode
				3x PWM mode (DEFAULT)
				3x direct PWM mode

Connections for Header J1-J2 on DRV8317H

Id.	Setting Name	Description	Position	Function
4	VIN_AVDD select	Use J1 to supply VIN_AVDD voltage from either VM or 5 V.		VIN_AVDD supplied from VM
				VIN_AVDD supplied from 5 V
5	HALL_EXT select	Use J2 to supply Hall power externally or from 3V3		Hall power supplied externally from the HALL PWR test point
				Hall power supplied from 3V3

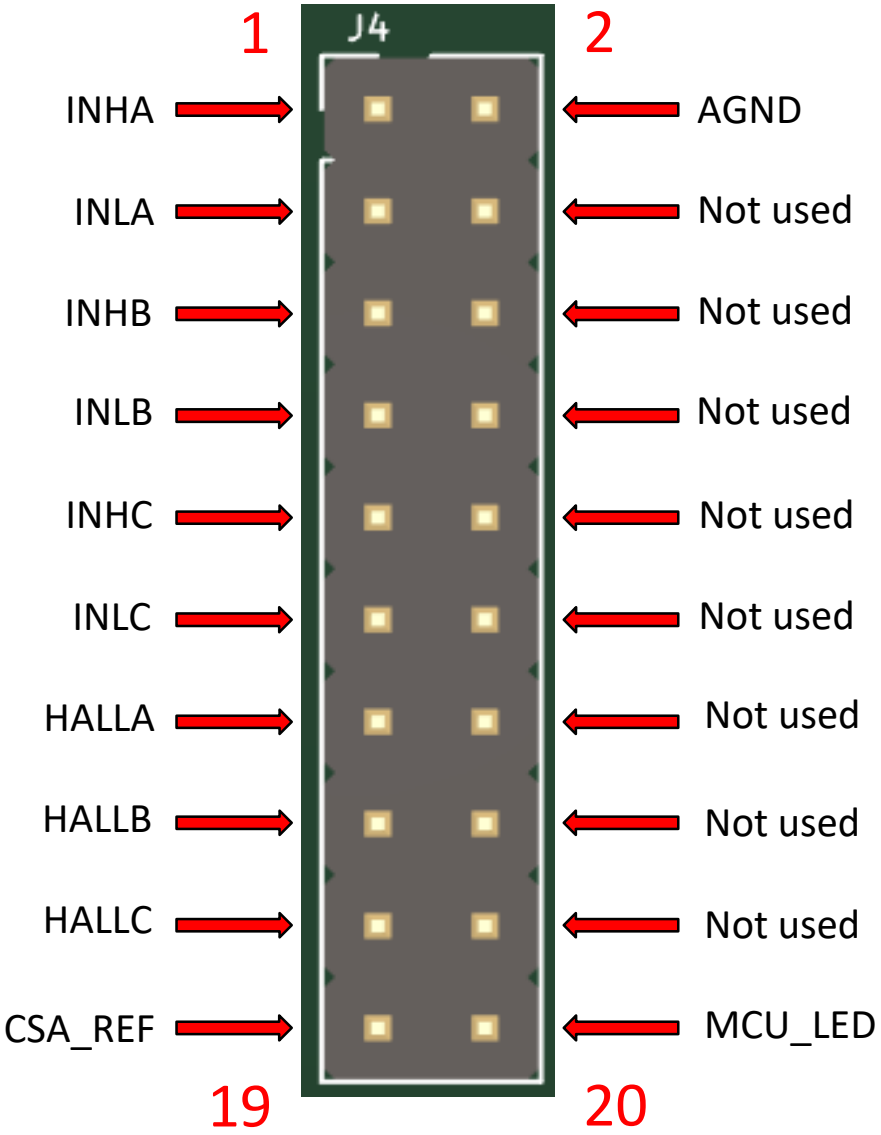
J3 PINOUTS for DRV8317H



Connections for Header J3 on DRV8317H

J3 Pin Number	DRV8317 Function	IO Connection Types	Description
1	3V3	POWER	3V3 LaunchPad Supply
2	5V	POWER	5 V Boost from LaunchPad
3	Not Used	Not Used	Not used
4	AGND	POWER	GND connection
5	Not Used	Not Used	Not Used
6	VSENA	ADC	Phase A Voltage Sense
7	Not Used	Not used	Not used
8	VSENB	ADC	Phase B Voltage Sense
9	nSLEEP	GPIO	For internal use only
10	VSENC	ADC	Phase C Voltage Sense
11	Not Used	Not used	Not used
12	VSENVM	ADC	VM Bus Voltage Sense
13	Not used	Not used	Not used
14	ISENA	ADC	Phase A low side Current Sense
15	nFAULT	GPIO	For internal use only
16	ISENB	ADC	Phase B low side Current Sense
17	nSLEEP	GPIO	Active-low input to sleep pin
18	ISENC	ADC	Phase C low side Current Sense
19	nFAULT	GPIO	Active-low output from fault pin
20	C_TAP	ADC	ADC for Center Tap sensing

J4 PINOUTS for DRV8317H



Connections for Header J4 on DRV8317H

J4 Pin Number	DRV8317 Function		Description
1	INHA	GPIO/PWM	PWM used to switch High-side FET of Phase A
2	AGND	POWER	GND connection
3	INLA	GPIO/PWM	PWM used to switch Low-side FET of Phase A
4	Not used	Not used	Not used
5	INHB	GPIO	PWM used to switch High-side FET of Phase B
6	Not used	Not used	Not used
7	INLB	GPIO/PWM	PWM used to switch Low-side FET of Phase B
8	Not used	Not used	Not used
9	INHC	GPIO/PWM	PWM used to switch High-side FET of Phase C
10	Not used	Not used	Not used
11	INLC	GPIO/PWM	PWM used to switch Low-side FET of Phase C
12	Not used	Not used	Not used
13	HALLA	GPIO	HALL sensor A from motor
14	Not used	Not used	Not used
15	HALLB	GPIO	HALL sensor B from motor
16	Not used	Not used	Not used
17	HALLC	GPIO	HALL sensor C from motor
18	Not used	Not used	Not used
19	CSA_REF	GPIO	For internal use only
20	MCU_LED	GPIO	Visual feedback for LaunchPad connection